Soluções - Folha 2

1a) $D_f = \mathbb{R}$; $D'_f =]-\infty, 4]$; $x = -2 \vee x = 2$

b) $D_f = \mathbb{R}$; $D'_f = [2, +\infty[$; Não tem zeros

c) $D_f = \mathbb{R}$; $D'_f = [-\frac{5}{9}, +\infty[$; $x = 1 \vee x = 2$

d) $D_f = \mathbb{R}$; $D'_f =]-\infty, \frac{9}{2}]$; $x = -2 \vee x = 1$

3a) $D_f =]-\infty, -3] \cup [2, +\infty[$

b) $D_f =]-\infty, 0[\cup [1, +\infty[$

c) $D_f = \mathbb{R} \setminus \{-3, 3\}$

d) $D_f = \mathbb{R}$

e) $D_f =]1, +\infty[$

f) $D_f =]-\infty, -3[\cup]3, +\infty[$

g) $D_f = [3, +\infty[$

h) $D_f = [-3, -1] \cup [1, 3]$

4a) $D_f = [0, +\infty[$; $D'_f = [-1, +\infty[$

b) $D_f =]-\infty, -1] \cup [1, +\infty[$; $D'_f = [0, +\infty[$

c) $D_f = [-2, +\infty[$; $D'_f =]-\infty, 5[$

d) $D_f = \mathbb{R} \setminus \{0\}$; $D'_f = \mathbb{R} \setminus \{0\}$

5 a) $f^{-1}(x) = \frac{1}{2-x}$; $D_{f^{-1}} = \mathbb{R} \setminus \{2\}$

b) $f^{-1}(x) = \frac{2x+1}{2-x}$; $D_{f^{-1}} = \mathbb{R} \setminus \{2\}$

c) $f^{-1}(x) = x^2 - 1$; $D_{f^{-1}} = [0, +\infty[$

d) $f^{-1}(x) = \frac{5x+3}{1-x}$; $D_{f^{-1}} = \mathbb{R} \setminus \{1\}$

6 a) $\frac{1}{4}$ b) $\frac{6}{37}$ c) 0 d) $\frac{x^2+x^3+x}{(x^2+1)^2}$

e) $\frac{x^2+x+1}{2}$ f) $\frac{x^2+x}{(x^2+x)^2+1}$

7 a) 12 b) $\frac{7}{2}$ c) 0

8 a) 3 b) $\frac{1}{4}$ c) $e \times 2^4$ d) 0 e) 4 f) 5

9 a) $x = \ln 2 \vee x = \ln 3$ b) $x = 0$ c) $x = \frac{7}{5}$

d) $x = 5/4$ e) $x = 0 \vee x = 2$

f) $x = 2 \vee x = 3$

10 a) $D_f = \mathbb{R}$ b) $D_f =]-1, +\infty[$

c) $D_f = \mathbb{R}$ d) $D_f =]-1, +\infty[\setminus \{0\}$

e) $D_f =]-\infty, -1[\cup]1, +\infty[$

10 f) $D_f = \mathbb{R} \setminus \{-1, 1\}$ g) $D_f = [-3, 0[\cup]0, 1[$

h) $D_f =]0, +\infty[\setminus \{1\}$ i) $D_f =]0, +\infty[$

j) $D_f = \mathbb{R} \setminus \{0\}$ k) $D_f =]4, 5] \cup]6, +\infty[$

13 a) $x = 1/2$

$D_f = \mathbb{R}$

$D'_f =]-\infty, 1[$

b) $x = 2 - e^{-2}$

$D_f =]-\infty, 2[$

$D'_f = \mathbb{R}$

14 a) $D_f = \mathbb{R} \setminus \{-1, 1\}$ e $D_g = [2, +\infty[$

b) $f \circ h :]-\infty, 1] \setminus \{0\} \longrightarrow \mathbb{R}$

$x \mapsto \frac{x-1}{x}$

15 a) $D_f = \mathbb{R}$ e $D'_f =]-1, +\infty[$

b) $f^{-1} :]-1, +\infty[\longrightarrow \mathbb{R}$

$x \mapsto -2 + \ln(x+1)$

16 a) $D_f =]-3, +\infty[$ e $D'_f = \mathbb{R}$

b) $f^{-1} : \mathbb{R} \longrightarrow]-3, +\infty[$

$x \mapsto -3 + e^{1-x}$

17 a) $D_f =]1, +\infty[$ e $D'_f = \mathbb{R}$

b) $f^{-1}: \mathbb{R} \longrightarrow]1, +\infty[$

$x \hookrightarrow 1 - 2^{x-1}$

18 a) $D_f = \mathbb{R} \setminus \{0\}$ e $D'_f =]-\infty, 1[\setminus \{-1\}$

b) $f^{-1}:]-\infty, 1[\setminus \{-1\} \longrightarrow \mathbb{R}$

$x \hookrightarrow \dfrac{1}{1 - \log_2 (1-x)}$

19 a) $D_f = \mathbb{R}$ e $D'_f =]-\infty, 1[$

b) condições impossíveis

20 a) $D_f = \mathbb{R}$ e $D'_f =]-1, e-1[$

b) $x = -1 \vee x = 1$